

Navigate through Enigmatic Labyrinth

A Survey of Chain of Thought Reasoning: Advances, Frontiers and Future

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Abstract

Reasoning, a fundamental cognitive process integral to human intelligence, has garnered substantial interest within artificial intelligence. Notably, recent studies have revealed that chain-of-thought prompting significantly enhances LLM’s reasoning capabilities, which attracts widespread attention from both academics and industry. In this paper, we systematically investigate relevant research, summarizing advanced methods through a meticulous taxonomy that offers novel perspectives. Moreover, we delve into the current frontiers and delineate the challenges and future directions, thereby shedding light on future research. Furthermore, we engage in a discussion about open questions. We hope this paper serves as an introduction for beginners and fosters future research. Resources have been made publicly available at <https://github.com/zchuz/CoT-Reasoning-Survey>.

1 Introduction

In the realm of human cognition, reasoning stands as the linchpin, essential in the understanding of the world and the formation of our decisions. As the scale of pre-training continues to expand (Brown et al., 2020; OpenAI, 2023; Touvron et al., 2023a,b), large language models (LLMs) exhibit growing capabilities in numerous downstream tasks (Wei et al., 2022a; Schaeffer et al., 2023; Zhou et al., 2023c). Recently, researchers have discovered that LLMs emerge with the capability for step-by-step reasoning through in-context learning, a phenomenon referred to as chain-of-thought (CoT) reasoning. It is broadly observed that CoT prompting significantly boosts the reasoning abilities of LLMs, especially in complex tasks (Wei et al., 2022b; Cobbe et al., 2021; Geva et al., 2021).

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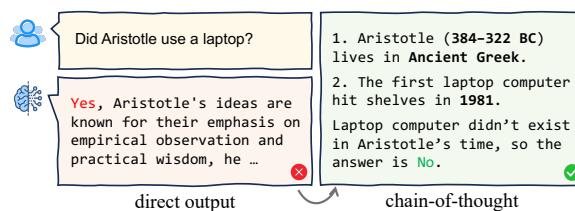


Figure 1: The model tackles complex problems step-by-step under the guidance of chain-of-thought prompting.

Figure 1 illustrates an example of chain-of-thought reasoning. Rather than directly providing the answer, chain-of-thought reasoning offers a step-by-step reasoning trajectory. Specifically, it decomposes intricate problems into manageable steps (*thoughts*), simplifying the overall reasoning process, and creates a linkage (*chain*) among the reasoning steps to ensure no important conditions are overlooked. Additionally, chain-of-thought reasoning offers an observable reasoning process, allowing users to comprehend the model’s decision-making trajectory and increase the trustworthiness and interpretability of the final answer.

Benefiting from the remarkable performance of CoT prompting, it has attracted widespread attention across both academia and industry, evolving into a distinct research branch within the field of prompt engineering (Liu et al., 2023d; Qiao et al., 2023). Moreover, it has emerged as a crucial component in the landscape of AI autonomous agents (Wang et al., 2023h; Xi et al., 2023). However, these studies still lack a systematic review and analysis. To fill this gap, we propose this work to conduct a comprehensive and detailed analysis of CoT reasoning. Specifically, this paper delves into the broader scope of chain-of-thought reasoning, which we refer to as generalized chain-of-thought (XoT). The core philosophy of XoT reasoning is the gradual unraveling of complex problems via a step-by-step reasoning approach.

Our contributions can be summarized as follows: (1) **Comprehensive Survey**: This is the first comprehensive survey dedicated for XoT reasoning; (2) **Meticulous taxonomy**: We introduce a meticulous taxonomy (shown in Figure 2); (3) **Frontier and Future**: We discuss new frontiers, outline their challenges, and shed light on future research. (4) **Resources**: We make the resources publicly available to facilitate the research community.

Survey Organization We first give background and preliminary (§2); then present benchmarks (§3) and advanced methods (§4) from different perspectives. Furthermore, we discuss frontier research (§5), and outline challenges as well as future directions (§6). Finally, we give a further discussion about open questions (§A.2).

2 Background and Preliminary

2.1 Background

Over the past few years, as the scale of pre-training continuously increases (Brown et al., 2020; Scao et al., 2022; Touvron et al., 2023b; Zhao et al., 2023b), language models have emerged with numerous new capabilities, such as in-context learning (Wei et al., 2022a; Brown et al., 2020) and chain-of-thought reasoning (Wei et al., 2022b). Accompanying this trend, pre-training then prompting has gradually replaced pre-training then fine-tuning as the new paradigm in natural language processing (Qiu et al., 2020; Zhao et al., 2023b).

2.2 Preliminary

In this section, we provide the preliminary for standard prompting and chain-of-thought reasoning. Referring to Qiao et al. (2023), we define the notations as follows: question Q , prompt $\mathcal{T}$, probabilistic language model p_{LM} and prediction $\mathcal{A}$.

First, we consider the few-shot standard prompting scenario, where prompt $\mathcal{T}_{SP}$ includes instruction I and few-shot demonstrations (several question-answer pairs). The model takes the question and prompt as inputs and produces the answer prediction $\mathcal{A}$ as its output, as shown in Equ. (1,2).

$$\mathcal{T}_{SP} = \{I, (x_1, y_1), \dots, (x_n, y_n)\} \quad (1)$$

$$p(\mathcal{A} | \mathcal{T}, Q) = \prod_{i=1}^{|\mathcal{A}|} p_{LM}(a_i | \mathcal{T}, Q, a_{<i}) \quad (2)$$

Next, we consider chain-of-thought prompting under few-shot setting, wherein the prompt $\mathcal{T}_{CoT}$

includes instruction, questions, answers, and rationales e_i . In chain-of-thought reasoning, the model no longer directly generates answers. Instead, it generates step-by-step reasoning trajectories $\mathcal{R}$ before giving answers $\mathcal{A}$, as shown in Equ. (3,4,5,6).

$$\mathcal{T}_{CoT} = \{I, (x_1, e_1, y_1), \dots, (x_n, e_n, y_n)\} \quad (3)$$

$$p(\mathcal{A}, \mathcal{R} | \mathcal{T}, Q) = p(\mathcal{A} | \mathcal{T}, Q, \mathcal{R}) \cdot p(\mathcal{R} | \mathcal{T}, Q) \quad (4)$$

$$p(\mathcal{R} | \mathcal{T}, Q) = \prod_{i=1}^{|\mathcal{R}|} p_{LM}(r_i | \mathcal{T}, Q, r_{<i}) \quad (5)$$

$$p(\mathcal{A} | \mathcal{T}, Q, \mathcal{R}) = \prod_{j=1}^{|\mathcal{A}|} p_{LM}(a_j | \mathcal{T}, Q, \mathcal{R}, a_{<j}) \quad (6)$$

2.3 Advantages of CoT Reasoning

As a novel reasoning paradigm, chain-of-thought gains various advantages. (1) Boosted Reasoning. Chain-of-thought reasoning breaks down complex problems into manageable steps and establishes connections among these steps, thereby facilitating reasoning. (2) Offering Interpretability. Chain-of-thought reasoning provides observable reasoning traces, allowing the user to understand the model’s decision, making the reasoning process transparent and trustworthy. (3) Advance Collaboration. Fine-grained reasoning traces facilitate user-system interaction, allowing for altering the model’s execution trajectory, thereby fostering the development of autonomous agents powered by LLMs.

3 Benchmarks

In this section, we briefly outline the benchmarks for evaluating reasoning capabilities, including mathematical, commonsense, symbolic, logical, and multi-modal reasoning. The overview of benchmarks is shown in Table 1. For more details about benchmarks, please refer to Appendix B.

Mathematical Reasoning Mathematical reasoning forms the foundation of human intelligence, playing a crucial role in problem-solving, decision-making, and world comprehension. It is commonly used to assess the general reasoning ability of LLMs (Patel et al., 2021; Cobbe et al., 2021; Hendrycks et al., 2021b; Mishra et al., 2022a).

Commonsense Reasoning Commonsense reasoning is essential for the interaction in daily life and the perception of the world, which assesses the world comprehension capacity of language models (Talmor et al., 2019, 2021; Geva et al., 2021).

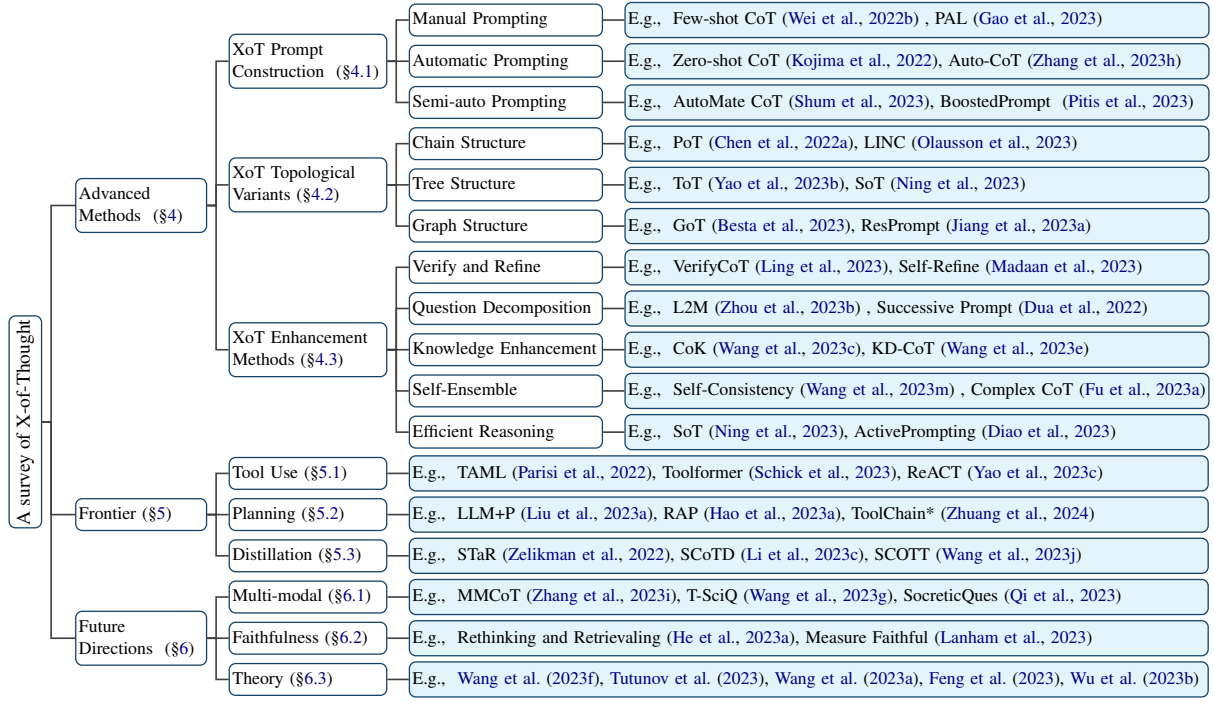


Figure 2: Taxonomy of Advanced Methods, Frontiers and Future Directions (Full version in Figure 8).

Symbolic Reasoning Symbolic reasoning disentangles semantics and serves as a testbed for language models’ competence in simulating atomic operations (Wei et al., 2022b; Srivastava et al., 2022; Suzgun et al., 2023).

Logical Reasoning Logical reasoning is of paramount importance as it serves as the bedrock for rational thinking, robust problem-solving and interpretable decision-making (Liu et al., 2020; Yu et al., 2020; Tafjord et al., 2021; Han et al., 2022).

Multi-modal Reasoning Multimodal reasoning seamlessly integrates textual thought with sensory experiences from the natural world, such as visual scenes, and auditory sounds, to create a richer, more comprehensive understanding of information (Zellers et al., 2019; Park et al., 2020; Xiao et al., 2021; Lu et al., 2022; Chen et al., 2023c).

4 Advanced Methods

This section discusses advanced XoT methods from three viewpoints: prompt construction (§4.1), topological variations (§4.2), and enhancement methods (§4.3). The taxonomy is shown in Figure 2.

4.1 XoT Prompt Construction

Based on the human effort for constructing chain-of-thought prompting, we divide the construction approaches into three categories: 1) Manual XoT, 2) Automatic XoT, and 3) Semi-automatic XoT.

4.1.1 Manual Prompting

Wei et al. (2022b) first proposes chain-of-thought prompting (Fewshot CoT) by manually annotating natural language form rationales to guide models in stepwise reasoning. Moreover, Fu et al. (2023a) discovers that using complex reasoning chains as demonstrations can further improve reasoning performance. Yet, the NL form reasoning encounters inconsistent reasoning. To mitigate intermediate errors in reasoning, PAL (Gao et al., 2023), PoT (Chen et al., 2022a), MathPrompter (Imani et al., 2023) and NLEP (Zhang et al., 2023d) leverage rationales in programming language form, transforming problem-solving into program generation, and obtaining a deterministic answer through external program executor. Although manual XoT demonstrates better performance, the annotation of rationales incurs a significant increase in cost and introduces dilemmas in demonstration selection.

4.1.2 Automatic Prompting

Some work designs specific instructions to stimulate CoT reasoning under zero-shot, such as appending *Let’s think step by step* after questions (Kojima et al., 2022). There are also other types of instructions, including writing programs to solve problems (Chen et al., 2022a), drafting plans before reasoning (Wang et al., 2023i), generating meta instructions based on task information (Crispino et al., 2023) and role playing (Kong et al., 2023a).

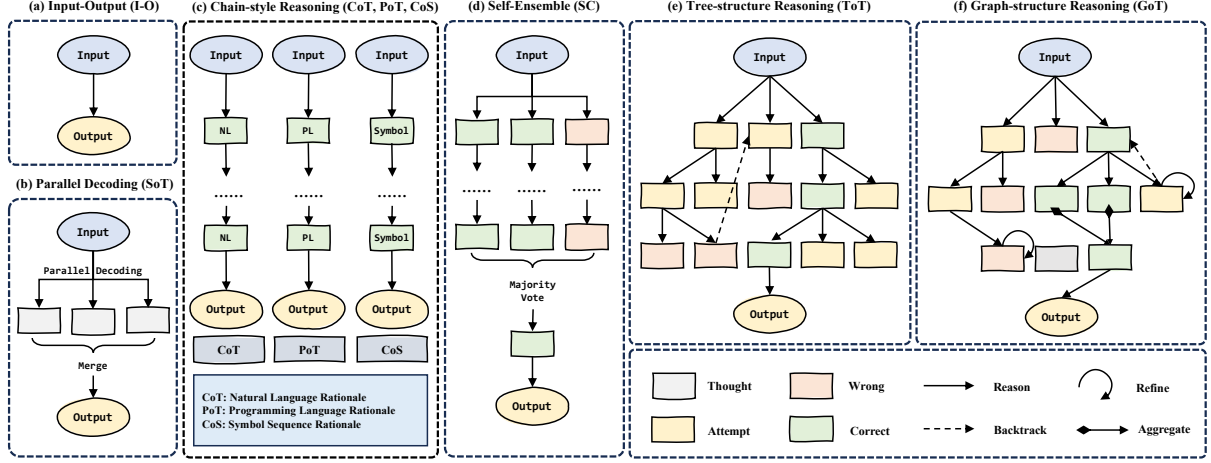


Figure 3: Topological variants emerging in the evolution of XoT. (a) standard I-O prompting, (b) parallel-constrained tree structure variants, (c) chain structure variants with distinct rationale descriptions, (d) chain structure variants with self-ensemble, (e) standard tree structure variants, and (f) standard graph structure variants.

However, due to the lack of guidance from clearly defined demonstrations, instruction-based methods appear extremely unstable. Another route of work conducts few-shot reasoning based on automatically generated rationales (usually by zero-shot CoT), which improves the stability of reasoning. These methods focus on selecting appropriate demonstrations. Zhang et al. (2023h) chooses diverse rationales through clustering, Zou et al. (2023) constructs demonstrations based on the question pattern, improving the generalization, Wan et al. (2023) employs answer entropy as a metric for selection, and Xu et al. (2023) uses Gibbs sampling to iteratively select demonstrations.

4.1.3 Semi-automatic Prompting

Building upon automatic XoT based on few-shot learning, semi-automatic approaches incorporate a small number of human-annotated rationales to obtain supervised signals. They focus on bootstrapping to acquire high-quality rationales and selecting appropriate demonstrations to facilitate reasoning. Shao et al. (2023b) generates high-quality rationales through alternating forward and backward synthetic processes, and Pitis et al. (2023) iteratively expands the examples when encountering challenging questions, which mitigates the issue of limited human supervision. On the other hand, some studies optimize demonstration selection. Shum et al. (2023) and Lu et al. (2023b) utilize policy gradient optimization to learn demonstration selection strategy, while Ye and Durrett (2023) searches the development set and selects proper demonstration using two proxy metrics.

4.1.4 Pros and Cons of Three Approaches

Manual prompting relies on high-quality rationale annotations, which result in better performance. However, it encounters drawbacks such as high labor costs and challenges in domain transfer. In contrast, automatic prompting incurs no labor costs and facilitates free domain transfer. However, it is plagued by errors and instability due to the absence of supervised signals. Semi-automatic prompting strikes a dedicated balance, achieving a trade-off between performance and costs, making it more suitable for downstream applications.

4.2 XoT Topological Variants

The evolution of XoT has led to the development of multiple topological variants¹. In this section, we will delve into topological variants of XoT: chain structure, tree structure and graph structure.

Chain Structure The description format of rationales significantly influences reasoning execution. PAL (Gao et al., 2023) and PoT (Chen et al., 2022a) use programming languages to depict the reasoning process, transforming problem-solving into code generation. Similarly, formal logic description languages are also used to depict logical reasoning (Olausson et al., 2023; Pan et al., 2023; Ye et al., 2023a). The aforementioned methods decouple the thought generation from execution, thereby eliminating inconsistency reasoning errors. Additionally, algorithmic descriptions (Sel et al., 2023) can offer a high-level reasoning framework

¹We consider XoT with chain structure and natural language rationales as vanilla CoT (the most primitive one).

instead of details, endowing the model with the ability for global thinking.

Tree Structure Chain structure inherently limits the scope of exploration. Through the incorporation of tree structures and search algorithms, models gain the capability to widely explore and backtrack during reasoning (Long, 2023; Yao et al., 2023b), as shown in Figure 3(e). Chen et al. (2024) iteratively explores and evaluates multiple tree-of-thoughts to further enhance reasoning. Benefiting from the exploration, tree variants have gained preliminary global planning capabilities towards the global optimum. Meanwhile, Mo and Xin (2023); Cao et al. (2023) introduce uncertainty measurement based on Monte Carlo dropout and generation likelihood, respectively, thereby offering a more accurate evaluation of intermediate reasoning processes. Yu et al. (2024) uses a bottom-up approach by building an analogy sub-problems tree. In addition, Ning et al. (2023) initially delivers reasoning drafts, accelerating reasoning by solving tree structure sub-problems in parallel. However, tree-based methods are restricted by demands of explicit question decomposition and state transition, which leads to limitations in task generalization.

Graph Structure Graph structures introduce loops and N-to-1 connections, enabling improved modeling of sub-problem aggregation and self-verification (Besta et al., 2023; Lei et al., 2023a), as illustrated in Figure 3(f). Graph structures outperform tree-based methods in handling complex problems. However, they rely on specially designed state decomposition, leading to poorer generalization. To address this, Jiang et al. (2023a) establishes an implicit graph upon the reasoning process through prompts, avoiding the constraints of explicit topological structures, thereby generalizing to various multi-step reasoning tasks.

The complex topological structure introduces a fine control flow, which facilitates LLMs in tackling harder problems. However, this complexity also limits the application of these methods in general reasoning, posing a significant challenge that needs to be addressed in future research.

4.3 XoT Enhancement Methods

This section introduces five enhanced XoT reasoning approaches, including verify and refine (§4.3.1), question decomposition (§4.3.2), knowledge enhancement (§4.3.3), self-ensemble (§4.3.4) and efficient reasoning (§4.3.5).

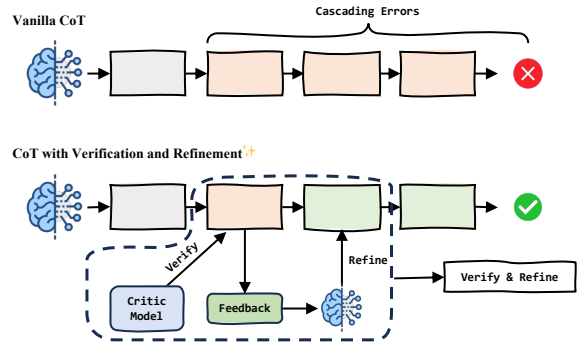


Figure 4: Verification and refinement rectify intermediate errors, which reduce cascading errors in reasoning.

4.3.1 Verify and Refine

LLMs tend to hallucinate, which manifests as factual and faithful errors in reasoning (Huang et al., 2023b). Incorporating verification and refinement can be an effective strategy for mitigating the phenomena. In this section, we primarily focus on mitigating faithful errors, with a separate discussion of factual errors in the following knowledge enhancement section (§4.3.3).

Reasoning can be refined based on critical feedback provided by LLMs. Paul et al. (2024a) trains a small critic model to provide structured feedback, but the quality of the feedback is limited due to the model size. Madaan et al. (2023) employs feedback from itself for iterative self-refinement, Li et al. (2023g) uses finer-grained feedback at the step level, and Shinn et al. (2023) further expands this method by incorporating long and short-term memory to provide more concise feedback. However, recent research suggests that LLMs may not address issues beyond their own capabilities (Kadavath et al., 2022; Yin et al., 2023), which raises doubt on the effectiveness of self-feedback (Huang et al., 2024a). To remedy this, some work incorporates external feedback (Gou et al., 2024a; Nathani et al., 2023) or performs secondary verification on the refined reasoning (Shridhar et al., 2023).

On the other hand, logical reasoning structures are also well-suited for verification. Ling et al. (2023) devises a deductive reasoning form named Natural Program, which guarantees that the conclusion is derived from the designated premises. Wu et al. (2024) applies a deductive filter to verify the entailment relationship between question and reasoning chains. Some studies perform step-wise verification during the beam search decoding stage. Xie et al. (2023) uses the log-probabilities of deduc-

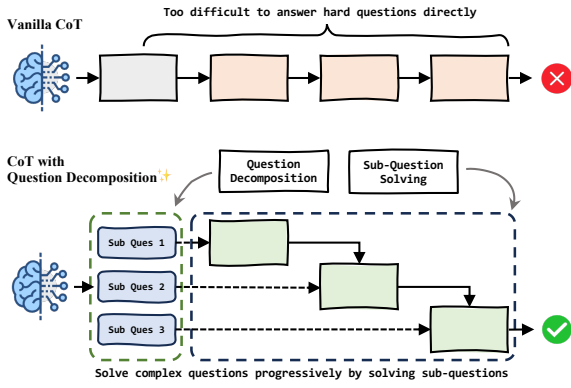


Figure 5: Question decomposition solves complex questions progressively by solving simple sub-questions.

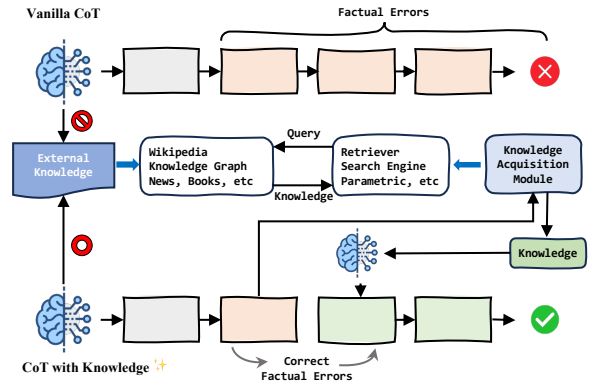


Figure 6: Incorporating knowledge (either internal or external) helps mitigate factual errors in reasoning.

tive reasoning as a search criterion, while [Zhu et al. \(2024a\)](#) trains a deductive discriminator for verification. Besides, backward (abductive) reasoning excels in detecting inconsistencies in reasoning. It reconstructs conditions or variables in the question based on the reasoning chain to discover inconsistencies, thereby refining the reasoning ([Xue et al., 2023](#); [Weng et al., 2022](#); [Jiang et al., 2023b](#)).

Reasoning with LLMs is prone to hallucinations, and feedback from intermediate steps plays a crucial role in refining the reasoning. However, the current acquisition of feedback signals still has many shortcomings, which necessitates further research.

4.3.2 Question Decomposition

The philosophy of XoT is to solve questions step-by-step. However, vanilla CoT does not explicitly decompose questions, making it challenging to answer complex questions. To address this, certain approaches address intricate problems by progressively tackling straightforward sub-problems.

L2M ([Zhou et al., 2023b](#)) initially breaks down the question into sub-questions in a top-down fashion. It then solves one sub-question at a time and leverages its solution to facilitate subsequent sub-questions. [Dua et al. \(2022\)](#) takes a similar approach to L2M, but it uses solutions from previous sub-questions to iteratively decompose questions. [Khot et al. \(2023\)](#) designs a modular task-sharing library that tailors more effective solutions to different classes of sub-questions. [Huang et al. \(2024b\)](#) breaks down the problem into a directed acyclic graph represented by QDMR, and then performs step-wise reasoning based on the graph dependencies. In multi-hop reasoning, iterative decomposition has become a common practice ([Wang et al., 2022](#); [Press et al., 2023](#); [Trivedi et al., 2023](#)). Ad-

ditionally, some methods obtain a dedicated decomposer through supervised training rather than relying on the LLM itself ([Li et al., 2023f](#); [Junbing et al., 2023](#)). However, when dealing with tabular reasoning, answering sub-questions may also pose a challenge, particularly when handling large tables. To tackle this issue, certain approaches involve decomposing both the questions and tables simultaneously ([Ye et al., 2023b](#); [Cheng et al., 2023](#); [Nahid and Rafiei, 2024](#)).

Bottom-up aggregation is also a viable solution, with a smaller exploration space. [Qi et al. \(2023\)](#) employs Socratic questioning for recursive self-questioning to solve complex questions, while [Zhang et al. \(2024\)](#), in a similar fashion, breaks down the conditions of complex problems into small components and resolves them bottom-up.

It should be noted that both decomposition and aggregation are highly dependent on the proper problem division, and reversely, a misaligned division may yield counterproductive results.

4.3.3 Knowledge Enhancement

When dealing with knowledge-sensitive tasks, LLMs often make factual errors. Introducing external knowledge or mining the model’s internal knowledge can help alleviate this issue. Some methods explicitly utilize the model’s intrinsic knowledge. For example, [Dhuliawala et al. \(2023\)](#); [Ji et al. \(2023\)](#); [Zheng et al. \(2024\)](#) prompt models to output its parametric knowledge, and then reason based on it. Additionally, [Zhang et al. \(2023f\)](#) prompts the model to perform inductive reasoning on its internal knowledge, deriving more general conclusions. Furthermore, [Liu et al. \(2023c\)](#) incorporates reinforcement learning to optimize introspective knowledge-grounded reasoning. Mean-

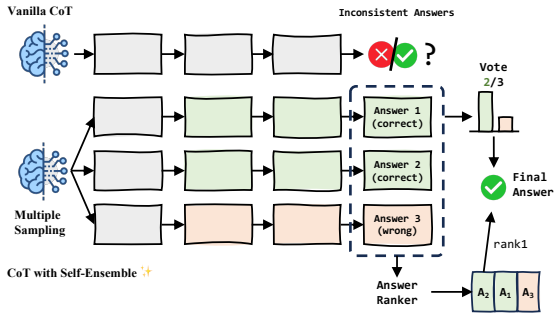


Figure 7: Self-ensemble reduces inconsistency by selecting final answers from multiple samplings.

while, [Li and Qiu \(2023\)](#) leverages model’s reasoning traces to construct a memory base, selecting relevant demonstrations whenever needed.

External knowledge is often more reliable than parametric knowledge. [Li et al. \(2023f\)](#); [Wang et al. \(2023e\)](#) generates queries based on the question, utilizing a knowledge base as the external knowledge. Building upon this, [Wang et al. \(2023c\)](#) introduces a verification step for the retrieved knowledge, further ensuring knowledge accuracy. However, when confronted with multi-hop reasoning, direct retrieval using the question can be insufficient. Therefore, [Press et al. \(2023\)](#); [Trivedi et al. \(2023\)](#); [Shao et al. \(2023a\)](#); [Yoran et al. \(2023\)](#) decompose the question and iteratively use sub-question for more precise retrieval.

4.3.4 Self-Ensemble

The sampling during generation introduces uncertainty, which in turn, creates the possibility of improving performance through self-ensemble. [Cobbe et al. \(2021\)](#) trains a verifier to rank answers, and [Hu et al. \(2024a\)](#) utilizes LLMs to self-rank their predictions. SC ([Wang et al., 2023m](#)) performs majority voting based on answers across multiple samples, and [Fu et al. \(2023a\)](#) proposes a complexity-based voting strategy on top of SC. Widespread practical evidence indicates that self-ensemble is an effective way to improve performance. However, answer-based ensemble fails to consider intermediate steps. In response, [Miao et al. \(2024\)](#); [Yoran et al. \(2023\)](#); [Khalifa et al. \(2023\)](#) refines the ensemble at the step level, and [Yin et al. \(2024\)](#) introduces hierarchical answer aggregation. Yet another concern is the limited diversity offered by probability sampling. To overcome this limitation, [Naik et al. \(2023\)](#) uses different instructions, [Liu et al. \(2023e\)](#) ensembles various XoT variants, and [Qin et al. \(2023\)](#) ensembles using multi-lingual

reasoning chains. Besides, the multi-agent debate (MAD) framework can also be regarded as heterogeneous ensemblings ([Liang et al., 2023](#); [Du et al., 2023](#); [Wang et al., 2023b](#)).

Self-ensemble, as a simple yet effective means, has gained widespread favor. Nevertheless, alongside the improvement in performance, there has been a multiplied increase in inference costs, which in turn limits its wide application.

4.3.5 Efficient Reasoning

LLMs are often inefficient in reasoning, such as high latency, substantial annotation costs, and elevated inference costs. To speed up reasoning, [Ning et al. \(2023\)](#) decomposes the questions in parallel and handles them simultaneously, [Zhang et al. \(2023b\)](#) generates a draft to skip intermediate layers during inference, and [Leviathan et al. \(2023\)](#); [Chen et al. \(2023a\)](#) introduce speculative decoding, which employs a smaller model for faster inference. [Diao et al. \(2023\)](#) annotates high-uncertainty samples to reduce human costs, and [Aggarwal et al. \(2023\)](#) dynamically adjusts sampling frequency to reduce inference costs. Further research should focus on efficient reasoning to promote the widespread application of LLMs.

5 Frontiers of Research

5.1 Tool Use

LLMs face difficulties in accessing news, performing calculations, and interacting with the environment. Previous work endows LLMs with the ability to use external tools, enhancing their reasoning capabilities and enabling them to interact with the (multi-modal) external environment ([Parisi et al., 2022](#); [Schick et al., 2023](#); [Shen et al., 2023a](#)).

However, these methods have limitations in facilitating multiple tool invocations and rectifying query errors. To tackle this problem, ReAct ([Yao et al., 2023c](#)) and Reflexion ([Shinn et al., 2023](#)) integrate the strengths of reasoning and action to complement each other. ART ([Paranjape et al., 2023](#)) uses a task library to select relevant tools and reasoning demonstrations. MM-REACT ([Yang et al., 2023b](#)) further incorporates vision experts to facilitate multi-modal reasoning and action.

Above-mentioned studies focus on leveraging external tools to grant LLMs the capacities they initially lacked, thereby improving their performance across various domains. Tool invocation facilitates interaction with external sources, enabling it to

gather additional information, while XoT enables effective elicitation, tracking, and action refining.

5.2 Planning

It is challenging for LLMs to provide accurate responses for complex goals, which requires planning to decompose them into sub-tasks and track the execution process. Plans can be described by code or definition languages. Sun et al. (2023) generates Python code to control the agent, and iteratively refine the plan based on the execution feedback. Liu et al. (2023a); Dagan et al. (2023) leverage the Planning Domain Definition Language (PDDL) (Gerevini, 2020) to describe the planning procedure. PDDL assists in decomposing complex problems and utilizing specialized models for planning before converting the results into natural languages. Zhou et al. (2023d) integrates self-refine (Madaan et al., 2023) with PDDL to achieve a better success rate in long-horizon sequential tasks.

Instead of pre-defined plans, many studies use search algorithms to dynamically plan and explore the action space. Tree-of-Thought explores the problem through DFS or BFS search, and tracks and updates the intermediate states (Yao et al., 2023b). RAP and LATS incorporate Monte Carlo Tree Search based on reasoning trajectories in planning (Hao et al., 2023a; Zhou et al., 2023a), and ToolChain* enables more efficient exploring through heuristic A* search (Zhuang et al., 2024).

LLMs, endowed with robust reasoning capabilities, can devise strategies for achieving complex goals. Furthermore, the integration of planning, reasoning, memory, and tool utilization serves as a cornerstone for LLM-powered autonomous agents.

5.3 Distillation of Reasoning Capabilities

In low-resource scenarios such as edge computing, distillation offers a possibility for deploying LLMs. Some methods employ self-distillation for self-improvement without external supervision. Huang et al. (2023a) employs self-consistency to generate reasoning chains from unlabeled data, followed by fine-tuning, enhancing its generalized reasoning capabilities. Zelikman et al. (2022) improves LM’s reasoning capabilities via self-loop bootstrapping.

Despite the powerful reasoning exhibited by CoT, it emerges primarily in large-scale LLMs, with its usage limited in smaller models. Magister et al. (2023) finds that smaller models, after fine-tuning on CoT reasoning data, can also exhibit the capacity for step-by-step reasoning. Fol-

lowing this trend, numerous studies attempt to distill the step-by-step reasoning capabilities of LLMs into smaller models. Hsieh et al. (2023b) employs self-consistency to filter predictions, distilling high-quality reasoning chains from LLMs. Ho et al. (2023); Li et al. (2023c) find that sampling multiple reasoning chains per instance is paramount for improving students’ reasoning capability. SCOTT (Wang et al., 2023j) utilizes contrastive decoding (Li et al., 2023e; O’Brien and Lewis, 2023) and counterfactual reasoning objective to tackle the shortcut problem. Li et al. (2024) improves the generalization of reasoning for unseen tasks through LoRA mixture-of-experts distillation.

Recent studies have found that the reasoning capabilities of small models can be further improved by optimizing over preference data. DialCoT (Han et al., 2023) decomposes reasoning steps into a multi-round dialog and optimizes the correct reasoning traces using PPO. Wang et al. (2023k); Feng et al. (2024) train a reward model on automatically generated data, which is designed to rank LLM’s reasoning traces, and then optimizes smaller models using PPO. (Xie et al., 2024) utilizes Monte Carlo Tree Search to sample and score reasoning trajectories, generates preference data on the fly, and uses DPO for online preference optimization.

Since code serves as an excellent intermediate representation for reasoning, Zhu et al. (2023) distills program-aided reasoning capability into smaller models. Meanwhile, some studies find that distilling reasoning chains from both natural language and code formats leads to further improvement (Li et al., 2023a; Zhu et al., 2024b). In addition to regular reasoning, Yang et al. (2024a) attempts to distill tabular reasoning capabilities, and Zhao et al. (2024b) seeks to endow smaller models with retrieval-augmented reasoning capabilities.

These studies adopt a shared paradigm that distills smaller models with reasoning chains generated from larger models with superior reasoning capabilities. However, it is worth noting that language models have intricate tradeoffs associated with multi-dimensional capabilities, and distilling task-specific reasoning ability may adversely downgrade the general performance (Fu et al., 2023b).

6 Future Directions

Despite XoT reasoning has showcased remarkable performance on numerous tasks, there are still some challenges that necessitate further research.

6.1 Multi-modal Reasoning

Current XoT research mostly focuses on plain text. However, interacting with the real world necessitates multi-modal capabilities. To facilitate research, SciQA (Lu et al., 2022) and CURE (Chen et al., 2023c) are introduced to emphasize multi-modal CoT reasoning. Through fine-tuning with the combination of vision and language features, Zhang et al. (2023i); Wang et al. (2023g) endow models with multi-modal CoT reasoning capabilities, and Yao et al. (2023d,a) further incorporate graph structures to model multi-hop relationships. Other approaches convert images to captions and use LLM for prompt-based reasoning (Yang et al., 2023b; Zheng et al., 2023b). However, the limited capabilities of vision-language models constrain their performance in multi-step reasoning (Alayrac et al., 2022; Li et al., 2023b; Peng et al., 2023).

Several critical challenges remain to be addressed in future research, which we summarize as follows: (1) Vision-text interaction: How can visual and textual features be effectively integrated, than solely depending on captions? (2) Harnessing VLLMs: How can we better apply LLM-based reasoning techniques to the multi-modal domain? (3) Video Reasoning: How to expand into video reasoning with complex temporal dependencies?

6.2 Faithful Reasoning

Extensive research indicates that LLMs often engage in unfaithful reasoning, such as factual errors and inconsistent reasoning. To address factual errors, one common approach is retrieval augmentation (Trivedi et al., 2023; Zhao et al., 2023a), but it requires appropriate timing and retrieval accuracy. Compared to factual errors, inconsistencies are more difficult to identify (Paul et al., 2024b). Common detection methods include deductive logic (Jiang et al., 2023b; Xue et al., 2023; Ling et al., 2023), post-processing (He et al., 2023a; Lei et al., 2023b), and critic-based approaches (Madaan et al., 2023; Nathani et al., 2023). Among them, Neural-symbolic reasoning (Chen et al., 2022a; Olausson et al., 2023) is a widely used approach for reducing inconsistencies, and question decomposition (Radhakrishnan et al., 2023) has also demonstrated its effectiveness to some degree. Furthermore, Zhang et al. (2023c); Lanham et al. (2023) investigate the factors influencing faithfulness from an empirical perspective.

Faithful reasoning encounters two significant

challenges: (1) Detection: How can unfaithful reasoning be accurately identified? (2) Correction: How can one obtain accurate feedback and make correct refinements based on that feedback?

6.3 Theoretical Perspective

The mechanism behind the CoT and ICL has not been clearly explained so far. Some studies empirically explore the roles of CoT and ICL in reasoning, offering practical insights (Wang et al., 2023a; Madaan and Yazdanbakhsh, 2022; Tang et al., 2023). Another line of work explores from a theoretical perspective. Li et al. (2023h); Feng et al. (2023); Merrill and Sabharwal (2023); Prystawski et al. (2023) investigate why CoT enhances reasoning abilities, while Wu et al. (2023b); Tutunov et al. (2023); Hou et al. (2023); Wang et al. (2023f) examine the mechanisms from a feature-based standpoint (information flow, attention, variables, etc.). Additionally, there have been preliminary explorations of the emergence mechanism (Schaeffer et al., 2023; Zhou et al., 2023c).

At present, the exploration of CoT theories is still limited to the surface level. There are still open questions that require further in-depth investigation. (1) How does the **emergence capability** arise? (2) **In what way** does CoT enhance reasoning compared to standard few-shot prompting?

7 Discussion

We delve into open questions about chain-of-thought reasoning, with the details discussion in Appendix A.2. The discussion encompasses three topics: (a) How does chain-of-thought reasoning ability emerge with large-scale pre-training? (b) How to provide accurate feedback for a model’s reasoning and decision-making. (c) The implications of chain-of-thought reasoning for LLM-powered autonomous agents and AGI.

8 Conclusion

In this paper, we conduct a systematic survey of existing research on generalized chain-of-thought reasoning, offering a comprehensive review of the field. Specifically, we meticulously categorize advanced methods, delve into current frontier research, highlight existing challenges, identify potential future research directions, and discuss open questions. This paper is the first systematic survey dedicated to CoT reasoning. We hope that this survey will facilitate further research in this area.

Limitations

This study provides the first comprehensive survey of generalized chain-of-thought (XoT) reasoning. Related work, benchmarks details and further discussion can be found in Appendix A,B.

We have made our best effort, but there may still be some limitations. On one hand, due to page limitations, we can only provide a brief summary of each method without exhaustive technical details. On the other hand, we primarily collect studies from *ACL, NeurIPS, ICLR, ICML, COLING and arXiv, and there is a chance that we may have missed some important work published in other venues. In the benchmarks section, we primarily list widely used datasets, and more complete benchmarks can be found in Guo et al. (2023). As of now, there is no definitive conclusion on open questions. We will stay abreast of discussions within the research community, updating opinions and supplementing overlooked work in the future.

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A Appendix

A.1 Related Survey

[Zhao et al. \(2023b\)](#) primarily focuses on the development of contemporary LLMs, while [Qiu et al. \(2020\)](#) surveys about early PLMs. Some works discuss reasoning in specific domains, such as mathematical reasoning ([Lu et al., 2023c](#)), common-sense reasoning ([Talmor et al., 2019](#)), and logical reasoning ([Yang et al., 2023c](#)). [Huang et al. \(2023b\)](#); [Zhang et al. \(2023e\)](#) conducts an investigation into potential hallucination phenomena in LLM’s reasoning. [Dong et al. \(2023\)](#) discusses in-context learning techniques in the era of LLMs, and [Yu et al. \(2023a\)](#) conducts a macroscopic investigation into natural language reasoning. [Liu et al. \(2023d\)](#) discusses prompt tuning, [Qiao et al. \(2023\)](#); [Yu et al. \(2023c\)](#); [Huang and Chang \(2023\)](#) focus on prompt engineering and reasoning strategies, and [Zhang et al. \(2023g\)](#) highlights the development from chain-of-thought reasoning to autonomous agents. This repository² also collects chain-of-thought reasoning papers.

Distinct from the above-mentioned surveys, this paper focuses on generalized chain-of-thought (XoT) reasoning in the era of LLMs. This is the first systematic investigation into XoT reasoning, and we hope our work can serve as an overview to facilitate future research.

A.2 Further Discussion

Open Question: Does CoT ability originate from code data pre-training? This is a pending question, initially summarized by [Fu and Khot \(2022\)](#) and widely circulated in the research community. In the early stages, LLMs like GPT3 ([Brown et al., 2020](#)) (davinci) and OPT ([Zhang et al., 2022b](#)) usually do not possess CoT capabilities, and they do not use or only incorporate a small amount of code data (not specialized) during pre-training. Recent models often incorporate specialized code data during pre-training, such as GPT-3.5, LLaMA2 ([Touvron et al., 2023b](#)) (with approximately 8% of code data during pre-training) and they all possess strong CoT capabilities. Additionally, [Gao et al. \(2023\)](#); [Chen et al. \(2022a\)](#) have found that the use of programming language form rationales can significantly enhance the model’s performance on complex reasoning tasks. Various indications point towards the source of CoT abilities lying in code data during pre-training.

²[Timothyxxx/Chain-of-ThoughtsPapers](#)

Recently, Ma et al. (2024) investigates the impact of code data on LLMs at different training stages, reaching the first qualitative conclusion supported by quantitative experimental results. They find that mixing code data during the pre-training stage enhances general reasoning abilities, while doing that in the instruction fine-tuning stage endows task-specific reasoning abilities.

Open Question: How to provide precise feedback on model’s reasoning or decisions? When dealing with multi-step reasoning or decision-making tasks, errors often occur in intermediate steps, and if these errors are not corrected promptly, they may lead to cascading errors. Currently, the primary methods for obtaining feedback include feedback from model itself (Madaan et al., 2023; Shinn et al., 2023), feedback from other models (Paul et al., 2024a), feedback from the external environment (Nathani et al., 2023; Gou et al., 2024a), and feedback based on reinforcement learning (Uesato et al., 2022; Lightman et al., 2024; Ma et al., 2023). However, some studies have raised doubts about the ability of LLMs to provide self-feedback (Huang et al., 2024a; Jiang et al., 2024). Generally speaking, certain issues exist in current methods. (1) How dependable is the feedback generated by the model itself? (2) Is there a fundamental distinction between feedback from other language models and self-feedback? (3) Does the feedback quality still remain constrained by the model’s capability boundaries? (4) How is external feedback for various scenarios pre-defined, and how can this be expanded to different scenarios?

In summary, there is currently no fully satisfying feedback approach and more research attention is needed on how to accurately obtain feedback signals from the model’s intermediate reasoning.

Discussion: Towards (early) AGI AGI has been the long-standing ultimate aspiration in the realm of artificial intelligence. Recent research on LLM-powered autonomous agents has successfully demonstrated a preliminary implementation of nascent artificial general intelligence (AGI).

Synergy between reasoning and interaction. Equipped with robust language comprehension capabilities, LLMs can interact with the external world through text-based interactions using plugins (tools, KB query, search engine, etc.) (Schick et al., 2023; Shen et al., 2023a; Qin et al., 2024). Combining powerful reasoning capabilities, LLMs have made significant strides in various planning

and decision-making tasks (Shinn et al., 2023; Yao et al., 2023b; Zhuang et al., 2024), catalyzing research on LLM-based autonomous agents (Wang et al., 2023h; Xi et al., 2023; Zhang et al., 2023g).

LLM acts as the Brain (Controller). In contrast to traditional AI, which concentrates on specific tasks, AGI seeks the ability to understand general tasks (Devlin et al., 2019; Dosovitskiy et al., 2021), covering a widespread spectrum. Within LLM-powered AI, the LLM typically serves as the brain (or central controller), handling reasoning, planning and decision-making, while delegating specific execution to dedicated modules (tools, weak AI, etc.) (Shen et al., 2023a; Yang et al., 2023a). LLM-powered AI has already diverged significantly from weak AI and is progressing toward human cognition and thinking.

While some studies suggest that LLMs represent an early manifestation of AGI (Bubeck et al., 2023; Jack, 2023), there are also scholars who contend that LLMs may not progress into AGI due to factors such as auto-regressive modeling and limited memory. As of now, there is still intense debate on whether LLMs can evolve into AGI. But regardless, LLM-powered AI has embarked on a distinctly different path from traditional AI, evolving towards a more generalized direction.

A.3 Early Attempts and Efforts in Specific Domains

In this section, we list the early attempts of XoT reasoning and efforts focused on specific domains.

Before the concept of CoT was introduced (Wei et al., 2022b), some efforts were made to enhance reasoning performance through the use of rationales (Marasovic et al., 2022; Rajani et al., 2019a,b; Dua et al., 2020). After that, certain work has empirically demonstrated the effectiveness of chain-of-thought prompting (Lampinen et al., 2022; Ye and Durrett, 2022; Arora et al., 2023) and Shi et al. (2023) explores multi-lingual CoT reasoning. Other work focuses on specific domains, such as machine translation (He et al., 2023b), sentiment analysis (Fei et al., 2023), sentence embeddings (Zhang et al., 2023a), summarization (Wang et al., 2023n), arithmetic (Lee and Kim, 2023), tabular reasoning (Chen, 2023; Ziqi and Lu, 2023), and backdoor attack (Xiang et al., 2024), etc. Katz et al. (2022); Zhang et al. (2022a) provide benchmarks and resources. Besides, some research utilizes specific pre-training to enhance reasoning (Lewkowycz et al., 2022; Zhao et al., 2022).

A.4 Empirical Results

We statistic the performance of various XoT methods in mathematics, commonsense, and symbolic reasoning, as shown in Table 2. We primarily collect the performance of GPT series models and the results are mainly from corresponding papers (some results are used as baselines in other papers). It is worth noting that due to variations in model checkpoints and experimental setups, even the methods with the same backbone LLM may not be fairly comparable. Therefore, this table only provides a rough trend of performance.

B Details of Benchmarks

B.1 Mathematical Reasoning

Mathematical reasoning is often used to measure the reasoning power of a model. Early benchmarks contain simple arithmetic operations (Hosseini et al., 2014; Koncel-Kedziorski et al., 2015; Roy and Roth, 2015; Koncel-Kedziorski et al., 2016). Ling et al. (2017) labels the reasoning process in natural language form, and Amini et al. (2019) builds on AQUA by labeling the reasoning process in program form. Later benchmarks (Miao et al., 2020; Patel et al., 2021; Cobbe et al., 2021; Gao et al., 2023) contain more complex and diverse questions. (Zhu et al., 2021; Chen et al., 2021, 2022b) require reasoning based on the table content. There are also competition-level benchmarks (Hendrycks et al., 2021b; Mishra et al., 2022a,b) and reading comprehension form benchmarks (Dua et al., 2019; Chen et al., 2023b).

B.2 Commonsense Reasoning

Commonsense reasoning entails the process of drawing inferences, forming judgments, and gaining insights based on widely known and commonly accepted world knowledge. Acquiring and understanding commonsense knowledge presents a significant challenge for models engaged in commonsense reasoning. Various benchmarks have been put forward to address these challenges, including commonsense understanding (Talmor et al., 2019, 2021; Bhakthavatsalam et al., 2021; Mihaylov et al., 2018; Geva et al., 2021; Huang et al., 2019; Bisk et al., 2020), event temporal commonsense reasoning (Rashkin et al., 2018; Zhou et al., 2019), and commonsense verification (Wang et al., 2019).

B.3 Symbolic Reasoning

Symbolic reasoning here refers specifically to the simulation of some simple operations, which are simple for humans yet challenging for LLMs. Last letter concatenation, coin flip, and reverse list (Wei et al., 2022b) are the most commonly used symbolic reasoning tasks. In addition, the collaborative benchmark BigBench (Srivastava et al., 2022) and BigBench-Hard (Suzgun et al., 2023) also contain several symbolic reasoning datasets, such as state tracking and object counting.

B.4 Logical Reasoning

Logical reasoning encompasses deductive reasoning, inductive reasoning, and abductive reasoning. Deductive reasoning derives conclusions from general premises (Liu et al., 2020; Yu et al., 2020; Tafjord et al., 2021; Han et al., 2022; Hong et al., 2023). Inductive reasoning derives general conclusions from special cases (Yang et al., 2024b). Abductive reasoning gives rational explanations for observed phenomena (Saparov and He, 2023).

B.5 Multi-modal Reasoning

In the real world, reasoning also involves information in modalities other than text, with visual modalities being the most prevalent. To this end, many benchmarks for visual multi-modal reasoning are proposed (Zellers et al., 2019; Park et al., 2020; Dong et al., 2022; Lu et al., 2022), and among them, ScienceQA (Lu et al., 2022) annotates reasoning process and is the most commonly used visual multi-modal reasoning benchmark. Video multi-modal reasoning (Lei et al., 2020; Yi et al., 2020; Wu et al., 2021; Xiao et al., 2021; Li et al., 2022; Gupta and Gupta, 2022) is more challenging as it introduces additional temporal information compared to visual multi-modal reasoning.

B.6 Comprehensive Benchmarks

Apart from the aforementioned individual datasets, there are also some comprehensive evaluation benchmarks. Some works aim to provide a holistic evaluation of the general reasoning capabilities (Srivastava et al., 2022; Suzgun et al., 2023; Hendrycks et al., 2021a; Huang et al., 2023d; Liang et al., 2022). In addition, there are also some multi-task benchmarks that focus on specific reasoning abilities, such as logical reasoning (Luo et al., 2023; Liu et al., 2023b) and temporal reasoning (Chu et al., 2023; Wang and Zhao, 2023).

Task	Dataset	Size	Input	Output	Rationale	Description
Mathematical Reasoning	AddSub (Hosseini et al., 2014)	395	Question	Number	Equation	Simple arithmetic
	SingleEq (Koncel-Kedziorski et al., 2015)	508	Question	Number	Equation	Simple arithmetic
	MultiArith (Roy and Roth, 2015)	600	Question	Number	Equation	Simple arithmetic
	MAWPS (Koncel-Kedziorski et al., 2016)	3,320	Question	Number	Equation	Simple arithmetic
	AQUA-RAT (Ling et al., 2017)	100,000	Question	Option	Natural Language	Math reasoning with NL rationale
	ASDiv (Miao et al., 2020)	2,305	Question	Number	Equation	Multi-step math reasoning
	SVAMP (Patel et al., 2021)	1,000	Question	Number	Equation	Multi-step math reasoning
	GSM8K (Cobbe et al., 2021)	8,792	Question	Number	Natural Language	Multi-step math reasoning
	GSM-Hard (Gao et al., 2023)	936	Question	Number	Natural Language	GSM8K with larger number
	MathQA (Amiri et al., 2019)	37,297	Question	Number	Operation	Annotated based on AQUA
	DROP (Dua et al., 2019)	96,567	Question+Passage	Number+Span	Equation	Reading comprehension form
	TheoremQA (Chen et al., 2023b)	800	Question+Theorem	Number	✗	Answer based on theorems
	TAT-QA (Zhu et al., 2021)	16,552	Question+Table+Text	Number+Span	Operation	Answer based on tables
	FinQA (Chen et al., 2021)	8,281	Question+Table+Text	Number	Operation	Answer based on tables
	ConvFinQA (Chen et al., 2022b)	3,892	Question+Table+Dialog	Number	Operation	Multi-turn dialogs
	MATH (Hendrycks et al., 2021b)	12,500	Question	Number	Natural Language	Challenging competition math problems
Commonsense Reasoning	NumGLUE (Mishra et al., 2022b)	101,835	Question+Text	Number+Span	✗	Multi-task benchmark
	LILA (Mishra et al., 2022a)	133,815	Question+Text	Free-form	Program	Multi-task benchmark
	ARC (Bhaktavatsalam et al., 2021)	7,787	Question	Option	✗	From science exam
	OpenBookQA (Mihaylov et al., 2018)	5,957	Question+Context	Option	✗	Open-book knowledges
	PIQA (Bisk et al., 2020)	21,000	Goal+Solution	Option	✗	Physical commonsense knowledge
	CommonsenseQA (Talmor et al., 2019)	12,247	Question	Option	✗	Derived from ConceptNet
	CommonsenseQA 2.0 (Talmor et al., 2021)	14,343	Question	Yes/No	✗	Gaming annotation with high quality
	Event2Mind (Rashkin et al., 2018)	25,000	Event	Intent+Reaction	✗	Intension commonsense reasoning
	McTaco (Zhou et al., 2019)	13,225	Question	Option	✗	Event temporal commonsense reasoning
	CosmosQA (Huang et al., 2019)	35,588	Question+Paragraph	Option	✗	Narrative commonsense reasoning
	ComValidation (Wang et al., 2019)	11,997	Statement	Option	✗	Commonsense verification
Symbolic Reasoning	ComExplanation (Wang et al., 2019)	11,997	Statement	Option/Free-form	✗	Commonsense explanation
	StrategyQA (Geva et al., 2021)	2,780	Question	Yes/No	✗	Multi-hop commonsense reasoning
	Last Letter Concat. (Wei et al., 2022b)	-	Words	Letters	✗	Rule-based
	Coin Flip (Wei et al., 2022b)	-	Statement	Yes/No	✗	Rule-based
	Reverse List (Wei et al., 2022b)	-	List	Reversed List	✗	Rule-based
Logical Reasoning	BigBench (Srivastava et al., 2022)	-	-	-	✗	Contains multiple symbolic reasoning datasets
	BigBench-Hard (Suzgun et al., 2023)	-	-	-	✗	Contains multiple symbolic reasoning datasets
	ReClor (Yu et al., 2020)	6,138	Question+Context	Option	✗	Questions from GMAT and LSAT
	LogiQA (Liu et al., 2020)	8,678	Question+Paragraph	Option	✗	Questions from China Civil Service Exam
	ProofWriter (Tafjord et al., 2021)	20,192	Question+Rule	Answer+Proof	Entailment Tree	Reasoning process generation
	FOLIO (Han et al., 2022)	1,435	Conclusion+Premise	Yes/No	✗	First-order logic
Multimodal Reasoning	DEER (Yang et al., 2024b)	1,200	Fact	Rule	✗	Inductive reasoning
	PrOntoQA (Saparov and He, 2023)	-	Question+Context	Yes/No+Process	First-Order Logic	Deductive reasoning
	VCR (Zellers et al., 2019)	264,720	Question+Image	Option	Natural Language	Visual commonsense reasoning
	VisualCOMET (Park et al., 2020)	1,465,704	Image+Event	Action+Intent	✗	Visual commonsense reasoning
	PMR (Dong et al., 2022)	15,360	Image+Background	Option	✗	Premise-based multi-modal reasoning
	ScienceQA (Lu et al., 2022)	21,208	Q+Image+Context	Option	Natural Language	Multi-modal reasoning with NL rationales
	VLEP (Lei et al., 2020)	28,726	Premise+Video	Option	✗	Video event prediction
	CLEVRER (Yi et al., 2020)	305,280	Question+Video	Option/Free-form	Program	Video temporal and causal reasoning
	STAR (Wu et al., 2021)	600,000	Question+Video	Option	✗	Video situated reasoning
	NEXT-QA (Xiao et al., 2021)	47,692	Question+Video	Option	✗	Video temporal, causal, commonsense reasoning
	Causal-VidQA (Li et al., 2022)	107,600	Question+Video	Free-form	Natural Language	Video causal and commonsense reasoning
	News-KVQA (Gupta and Gupta, 2022)	1,041,352	Q+V+KG	Option	✗	Video reasoning with external knowledge

Table 1: An overview of benchmarks and tasks on reasoning.

Method	Setting	Backbone	Mathematical				Commonsense		Symbolic	
			GSM8K	SVAMP	Asdiv	AQuA	CSQA	StrategyQA	LastLetterConcat	CoinFlip
I-O Prompting (Brown et al., 2020)	fewshot	text-davinci-002	19.7	69.9	74	29.5	79.5	65.9	5.8	49.0
Fewshot CoT (Wei et al., 2022b)	fewshot	text-davinci-002	63.1	76.4	80.4	45.3	73.5	65.4	77.5	99.6
PoT (Chen et al., 2022a)	fewshot	text-davinci-002	80	89.1	-	58.6	-	-	-	-
Complex CoT (Fu et al., 2023a)	fewshot	text-davinci-002	72.6	-	-	-	-	77	-	-
Automate CoT (Shum et al., 2023)	fewshot	text-davinci-002	49.7	73.3	74.2	37.9	76.1	67.9	58.9	-
Fewshot CoT (Wei et al., 2022b)	fewshot	text-davinci-003	66.83	69.06	-	29.13	-	-	-	-
PHP (Zheng et al., 2023a)	fewshot	text-davinci-003	79	84.7	-	58.6	-	-	-	-
Self-consistency (Wang et al., 2023m)	fewshot	text-davinci-003	67.93	83.11	-	55.12	-	-	-	-
Active Prompt (Diao et al., 2023)	fewshot	text-davinci-003	65.6	80.5	79.8	48	78.9	74.2	71.2	-
Synthetic Prompt (Shao et al., 2023b)	fewshot	text-davinci-003	73.9	81.8	80.7	-	-	-	-	-
FOBAR (Jiang et al., 2023b)	fewshot	text-davinci-003	79.5	86	-	58.66	-	-	-	-
Boosted Prompting (Pitis et al., 2023)	fewshot	text-davinci-003	71.6	-	-	55.1	-	-	-	-
Fewshot CoT (Wei et al., 2022b)	fewshot	code-davinci-002	60.1	75.8	80.1	39.8	79	73.4	70.4	99
Self-Consistency (Wang et al., 2023m)	fewshot	code-davinci-002	78	86.8	87.8	52	81.5	79.8	73.4	99.5
PAL (Gao et al., 2023)	fewshot	code-davinci-002	72	79.4	79.6	-	-	-	-	-
Resprompt (Jiang et al., 2023a)	fewshot	code-davinci-002	66.6	-	-	45.3	-	-	-	-
DIVERSE (Li et al., 2023g)	fewshot	code-davinci-002	82.3	87	88.7	-	79.9	78.6	-	-
Least-to-Most (Zhou et al., 2023b)	fewshot	code-davinci-002	68.01	-	-	-	-	-	94	-
Boosted Prompting (Pitis et al., 2023)	fewshot	code-davinci-002	83.3	88.6	-	61.7	-	-	-	-
Fewshot CoT (Wei et al., 2022b)	fewshot	gpt-3.5-turbo	76.5	81.9	-	54.3	78	63.7	73.2	99
Self-consistency (Wang et al., 2023m)	fewshot	gpt-3.5-turbo	81.9	86.4	-	62.6	-	-	-	-
MetaCoT (Zou et al., 2023)	fewshot	gpt-3.5-turbo	75.1	88.6	-	54.7	72.4	64.5	77.2	100
Verify CoT (Ling et al., 2023)	fewshot	gpt-3.5-turbo	86	-	-	69.5	-	-	92.6	-
Active Prompting (Diao et al., 2023)	fewshot	gpt-3.5-turbo	81.8	82.5	87.9	55.3	-	-	-	-
RCoT (Xue et al., 2023)	fewshot	gpt-3.5-turbo	84.6	84.9	89.3	57.1	-	-	-	-
FOBAR (Jiang et al., 2023b)	fewshot	gpt-3.5-turbo	87.4	87.4	-	57.5	-	-	-	-
Memory-of-Thought (Li and Qiu, 2023)	fewshot	gpt-3.5-turbo	-	-	-	54.1	-	-	-	-
Adaptive-consistency (Aggarwal et al., 2023)	fewshot	gpt-3.5-turbo	82.7	85	83	-	-	67.9	-	-
Boosted Prompting (Pitis et al., 2023)	fewshot	gpt-3.5-turbo	87.1	-	-	72.8	-	-	-	-
Zeroshot CoT (Kojima et al., 2022)	zeroshot	text-davinci-002	40.5	63.7	-	31.9	64	52.3	57.6	87.8
PoT (Chen et al., 2022a)	zeroshot	text-davinci-002	57	70.8	-	43.9	-	-	-	-
AutoCoT (Zhang et al., 2023h)	zeroshot	text-davinci-002	47.9	69.5	-	36.5	74.4	65.4	59.7	99.9
COSP (Aggarwal et al., 2023)	zeroshot	code-davinci-001	8.7	-	-	-	55.4	52.8	-	-
Plan-and-Solve (Wang et al., 2023i)	zeroshot	text-davinci-003	58.2	72	-	42.5	65.2	63.8	64.8	96.8
Agent-Instruct (Crispino et al., 2023)	zeroshot	gpt-3.5-turbo	73.4	80.8	-	57.9	74.1	69	99.8	95.2
Self-Refine (Madaan et al., 2023)	zeroshot	gpt-3.5-turbo	64.1	-	-	-	-	-	-	-
RCoT (Xue et al., 2023)	zeroshot	gpt-3.5-turbo	82	79.6	86	55.5	-	-	-	-

Table 2: The performance of various XoT methods in commonly used mathematical, commonsense and symbolic reasoning benchmarks. It is worth noting that, due to variations in the experimental setups of different methods, their performances are not directly comparable. The table is used to provide an overall empirical insight.

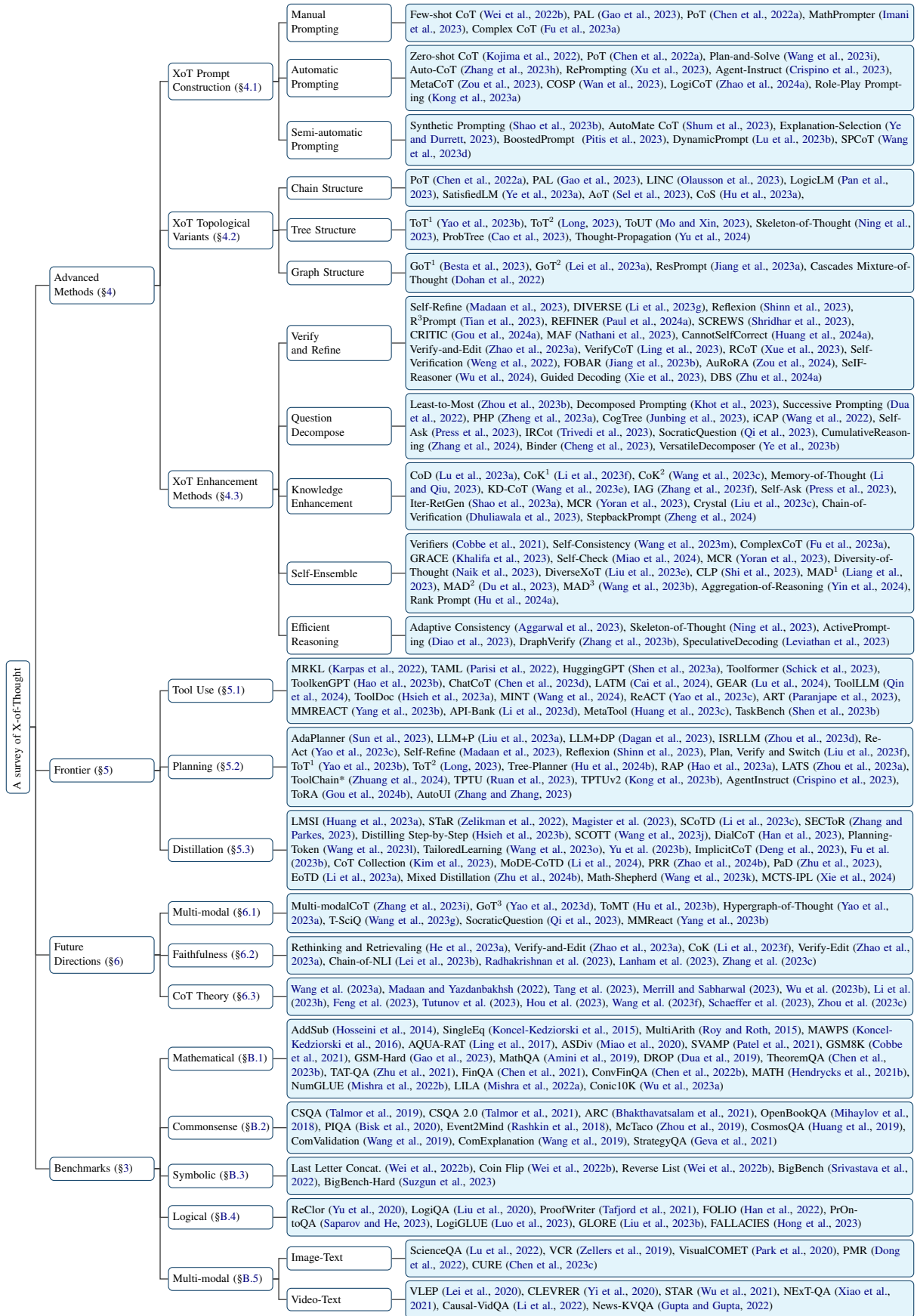


Figure 8: Taxonomy of Advanced Methods, Frontiers, Future Directions, and Benchmarks (Full Edition).